



UNSW
SYDNEY

COMP4141

Theory of Computation

Mid-Course Recap

Outline

Week 1: Formal Languages

Weeks 2-3: Regular Languages

Week 4: Context-free Languages

Week 5: Recursive and Recursively Enumerable Languages

Week 1 recap

Formal languages

- Alphabets, Strings/words, Languages
- Recursive definitions

Need to know

- Recursive definitions
- Induction proofs

Example problem

Example

Let Σ be a finite set.

Formally define the function $\text{rev} : \Sigma^* \rightarrow \Sigma^*$ which reverses the symbols of the word. For example $\text{rev}(abc) = cba$.

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Formally define the function $\text{rev} : \Sigma^* \rightarrow \Sigma^*$ which reverses the symbols of the word. For example $\text{rev}(abc) = cba$.

- $\text{rev}(\epsilon) = \epsilon$
- For $x \in \Sigma$ and $w \in \Sigma^*$: $\text{rev}(xw) = \text{rev}(w)x$

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Weeks 2-3 recap

Regular Languages

- Deterministic Finite Automata (DFA)
 - $(Q, \Sigma, \delta, q_0, F)$
 - $L(A)$
- Non-deterministic Finite Automata (NFA)
 - Non-determinism
 - ϵ -transitions
- Equivalence of DFAs and NFAs
 - Subset construction
- Regular expressions
 - Definition, $L(R)$
 - Equivalence to DFAs
- Pumping Lemma
- Myhill-Nerode theorem

Need to know

- DFAs, NFAs, Regular Expressions: Definitions, how to use
 - For example: Finding A or L such that $L = L(A)$
- Arguing correctness (formal vs informal)
- Pumping Lemma - how to use it, limitations
- Myhill-Nerode theorem - how to use it, limitations

Example problems

Example

Are the following languages regular or not? Justify your answer.

a $\{0^n 1^m : n - m = 2023\}$

b $\{0^n 1^m : n - m \geq 2023\}$

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Week 4 recap

Context-free languages

- Context-free grammars
 - Derivations, parse trees, ambiguity
- Pushdown automata
- Pumping lemma for CFLs
- Closure properties of CFLs
- Chomsky normal form
- Non-emptiness and word acceptance algorithms
 - CYK algorithm
- Chomsky hierarchy

Need to know

- CFGs and PDAs: Definitions, how to use
- Pumping lemma: How to use
- Algorithms for CFLs
 - Test for non-emptiness
 - Convert to Chomsky NF
 - CYK algorithm

Example problems

Example

Are the following languages context-free or not? Justify your answer.

a $\{0^n 1^m : n - m = 2023\}$

b $\{0^n 1^m 0^k : nm = k\}$

Example problems

Example

Let $\Sigma = \{0, 1\}$ and consider the following context-free grammar:
 $G = (V, \Sigma, R, S)$ where:

- $V = \{S, T\}$
- R is defined as:

$$\begin{array}{lcl} S & \rightarrow & 0T \mid T0 \mid \epsilon \\ T & \rightarrow & 1S \mid S1 \end{array}$$

and the following regular expression:

$$E = (01 \cup 10)^*$$

- a** Prove that $L(G) \not\subseteq L(E)$
- b** Prove that $L(E) \not\subseteq L(G)$

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Week 5 recap

Turing Machines, Recursive and Recursively Enumerable languages

- Turing Machines
 - Definitions, accept vs decide
 - Variants: stay-put, semi-infinite tape, multi-tape, non-determinism
 - Enumerators
 - Universal Turing Machine
- **Rec, RE, coRE**
 - **$\text{Rec} = \text{RE} \cap \text{coRE}$**
- Encodings
- Reductions: many-one vs Turing
- Undecidable problems: $A_{TM} = \{\langle M, w \rangle : M \text{ accepts } w\}$
- (Rice's theorem)

Need to know

- TMs: Definitions, how they work
- Many-one reductions: Definitions, how to use
- (Oracles)
- (Rice's theorem: When it applies)

Example problems

Example

A function $f : \Sigma^* \rightarrow \Gamma^*$ is **computable** if there is a Turing Machine M which, on input $x \in \Sigma^*$ halts with $f(x)$ on the tape. Prove that $f : \Sigma^* \rightarrow \Gamma^*$ is computable if and only if

$$L_f = \{\langle x, f(x) \rangle : x \in \Sigma^*\}$$

is recursively enumerable.

Example problems

Example

Show that the following language is not in **coRE**:

Input: A Turing Machine M

Question: Does M halt in the reject state on input ϵ ?